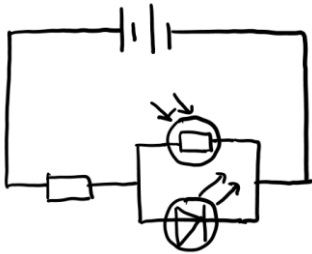


6(a)	By the principle of potential divider, $V_{\text{voltmeter}} = V_{\text{total}} \frac{R_{\text{voltmeter}}}{R_{\text{total}}} = \left(\frac{11000}{11000 + 1000} \right) 12$ $V_{\text{voltmeter}} = 11 \text{ V}$	M1 A1
(b)(i)	$I = \frac{3.0}{11000} = 2.7 \times 10^{-4} \text{ A}$	A1
(b)(ii)	Since 9 V must exist across CJ, by principle of potential divider, the contact should be 0.75 x the length of CD = 0.75 m.	A1
(b)(iii)	$R = \frac{V^2}{P} = \frac{3.0^2}{0.6}$ $= 15 \Omega$	M1 A1
(c)	The lamp has a much smaller resistance compared to the voltmeter, resulting in a much smaller effective resistance across JD. Hence, potential difference across JD will be much smaller than 3.0 V, insufficient to provide a noticeable brightness in the lamp. <i>Note: A lamp of 15 Ω in parallel with a resistance of 250 Ω gives an effective resistance of 14 Ω. The potential difference across the lamp is then only $\frac{14}{14+750} \times 12 = 0.22 \text{ V}$, which is much lesser than the rating of 3.0 V.)</i>	B1 B1

(d)



B1 - LED is forward biased and in parallel with LDR

B1 - fixed resistor in series